

Geometry
Unit 1
Lesson 11 Practice

Name ___ **Answer Key** ___
Date ___
Period ___

Steps to construct the perpendicular bisector of $\overline{HM}$ are shown out of order.

Step	Description
A	Repeat Steps 1 and 2 to create an arc centered at point M .
B	Label the point of intersection of $\overline{TW}$ and $\overline{HM}$ as R .
C	Without changing the width of the compass, draw an arc above and below the line centered at point H .
D	Draw $\overline{TW}$.
E	Set the compass point at point H and open the compass to a width approximately two-thirds the length of $\overline{HM}$.
F	Mark the two points of intersections of the arcs as T and W .

Use the letter of the steps to correctly reorder the steps for constructing a perpendicular bisector.

	Letter
1.	<i>E</i>
2.	<i>C</i>
3.	<i>A</i>
4.	<i>F</i>
5.	<i>D</i>
6.	<i>B</i>

$\overline{YK}$ is a perpendicular bisector of $\overline{HB}$. $HY = 7x + 13$, $HW = 8y - 15$, $WB = 3y + 21$, and $YB = 12x - 12$.

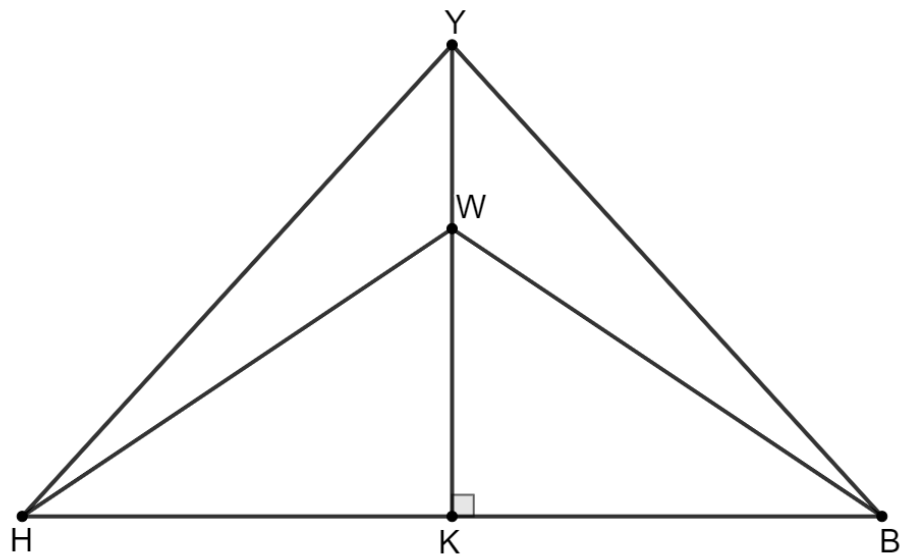
7. Determine the value of x .

$$7x + 13 = 12x - 12$$

$$13 = 5x - 12$$

$$25 = 5x$$

$$5 = x$$



8. Determine the length of $\overline{HY}$.

$$7(5) + 13 = 48$$

9. Determine the value of y .

$$8y - 15 = 3y + 21$$

$$5y - 15 = 21$$

$$5y = 36$$

$$y = 7.2$$

10. Determine the length of $\overline{WB}$.

$$3(7.2) + 21 = 42.6$$